Technical Report: Photonic Integrated Circuit (PIC) Layout Analysis

1. Executive Summary This report provides a comprehensive layout and functional analysis of the fabricated Photonic Integrated Circuit (PIC). The design integrates an array of four Mach-Zehnder Interferometers (MZIs) featuring progressive path length differences (ΔL), alongside a distributed contra-directional coupler (CDC) structure. This architecture is typically implemented for wavelength filtering, optical spectrum analysis, dispersion characterization, and narrowband reflection filtering in silicon-on-insulator (SOI) or similar integrated photonic platforms.
2. Circuit Architecture and Layout Description The chip layout comprises distinct functional blocks interconnected via single-mode rib or strip waveguides, interfaced through standardized optical input/output structures:

Optical I/O Interfaces: Edge or grating couplers (visible as tapered fan-out structures with grating teeth) facilitate efficient optical coupling between single-mode optical fibers and the on-chip waveguide network.

MZI Array (4 Channels): Four parallel Mach-Zehnder Interferometers configured with systematically scaled arm length differences (ΔL). Each MZI consists of an input splitter (e.g., a multi-mode interference [MMI] coupler or directional coupler), two arms with an engineered path asymmetry (ΔL), and an output combiner.

MZI 1: Minimal path delay (ΔL 1

), yielding a wide Free Spectral Range (FSR).

MZI 2 & MZI 3: Intermediate path delays (ΔL 2 <ΔL 3

), providing progressively narrower spectral periodicities.

MZI 4: Maximum path delay (ΔL 4

), characterized by extended serpentine routing to accommodate a large optical path difference for high-resolution spectral filtering.

Contra-Directional Coupler (CDC): Positioned on the rightmost section of the layout, this long, vertically oriented coupled-waveguide structure incorporates periodic Bragg corrugations along the interaction length. It acts as a contra-directional coupler for optical add-drop or narrow-band reflection filtering.

3. Theoretical Principles and Operation 3.1 MZI Array with Variable Path Lengths (ΔL) The transmission response of an asymmetric MZI is governed by the optical phase difference accumulated between its two arms:

$\Delta\phi = \lambda\ 2\pi\ \ n\ eff\ \ \Delta L$ where n eff

is the effective refractive index of the waveguide mode, λ is the operating wavelength, and ΔL is the physical path length difference.

The Free Spectral Range (FSR) of each MZI is inversely proportional to its path delay:

FSR≈ n g ΔL λ 2

where n g

is the group index. By cascading or testing four MZIs with distinct ΔL values, the circuit allows for multi-scale spectral characterization, vernier effect measurements, or tunable filtering across different channel spacing requirements.

3.2 Contra-Directional Coupler (CDC) The contra-directional coupler utilizes grating-assisted coupling between two parallel waveguides. Periodic perturbations along the waveguides satisfy the phase-matching condition for backward coupling:

β 1 +β 2

Λ 2π

where β 1

and β 2

are the propagation constants of the coupled modes, and Λ is the grating period. This structure reflects a narrow band of wavelengths backward into the drop port while transmitting the rest of the spectrum forward, providing high out-of-band rejection and a compact footprint ideal for wavelength division multiplexing (WDM).

4. Simulation and Design Verification Workflow To validate and optimize this PIC design using standard photonics computer-aided design (PCAD) tools (such as Ansys Lumerical or KLayout):

Mode Analysis: Use finite-difference eigenmode (FDE) solvers (e.g., Lumerical MODE) to determine single-mode dimensions, dispersion profiles, and group indices (n g

) for the chosen etch depth and waveguide width.

Component-Level Simulation:

Simulate the MMI splitters/combiners and waveguide bends using 3D Finite-Difference Time-Domain (FDTD) or varFDTD to minimize insertion loss and back-reflections.

Simulate the grating-assisted CDC to extract the reflection bandwidth, central Bragg wavelength, and coupling coefficient (κ).

Circuit-Level Netlist Simulation: Implement compact models in Lumerical INTERCONNECT to simulate the composite spectral response of the four-MZI array and the CDC under varying fabrication process corners.

5. Applications Wavelength Division Multiplexing (WDM): Channel dropping and routing using the sharp spectral response of the CDC.

Optical Sensing: High-sensitivity refractive index or temperature sensing leveraging the high-resolution interference fringes of the large- ΔL MZI.

On-Chip Spectrometry: Reconstructing broadband optical spectra using multi-FSR interferometric decoding.